The geometry of paths seen through bounded linear systems

Response distance and tensor certificates

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Abstract

How different do two driving paths look to linear systems with bounded matrix coefficients? This discussion draft measures their separation by worst-case terminal-response difference and compares it with a tensor certificate: a weighted Hilbert–Schmidt signature difference that bounds response distance from above. Two structured comparisons quantify the gap. For planar bounded-variation paths and their degree-two log-ODE approximations, a fixed partition with nonzero cubic discrepancy gives small-amplitude ratio $2\sqrt{2}/9$, uniformly in response dimension at least two. For chains of at least three adjacent area-only blocks, explicit separation persists at every positive amplitude, with bounds independent of chain length.

1 Response distance and tensor geometry

We ask how bounded linear systems distinguish paths, and how response distance compares with tensor-norm differences between their signatures or approximations.

Response distance. Let $X, \widetilde{X} : [0,1] \to \mathbb{R}^d$ be continuous paths of finite Euclidean variation, defined for X by

$$\|X\|_{1\text{-var}} = \sup_{\tau} \sum_{[u,v]\in\tau} \|X_v - X_u\|_2 \,,$$

where τ ranges over finite partitions into intervals and $\|\cdot\|_2$ is Euclidean norm. BV abbreviates bounded variation. For a tuple $A = (A_1, \dots, A_d)$ of real $n \times n$ matrices, let $\Phi^A(X) = Y_1$ be terminal response of

$$dY = \sum_i Y A_i \, dX^i, \qquad Y_0 = I_n,$$

where I_n is identity. Input dimension is d; response dimension is n. Fix joint coefficient budget $|A|^2 = \sum_i \|A_i\|_F^2 \le r^2$, $r > 0$, using Frobenius norm $\|A_i\|_F^2 = \operatorname{tr}(A_i^\top A_i)$. Use Euclidean operator norm $\|M\|_{\mathrm{op}} = \sup_{\|v\|_2=1} \|Mv\|_2$. Define

$$\Delta_{r,n}(X, \widetilde{X}) = \sup_{|A|\le r} \left\| \Phi^A(X) - \Phi^A(\widetilde{X}) \right\|_{\mathrm{op}} . \tag{1}$$

This measures largest terminal-response difference over admissible systems and unit initial states. It is a pseudodistance: distinct paths can have identical responses and hence zero separation. Requiring separation at most $\varepsilon > 0$ expresses response accuracy in (ε, r)-representation framework [1].

Tensor certificate. The signature records ordered iterated integrals:

$$S(X)_0 = 1, \qquad S(X)_j = \int_{0<u_1<\cdots<u_j<1} dX_{u_1} \otimes \cdots \otimes dX_{u_j}, \quad j \ge 1.$$

For standard basis (e_i), tensor word $e_{i_1} \otimes \cdots \otimes e_{i_j}$ records a sequence of j directions. Define $R_A(e_{i_1} \otimes \cdots \otimes e_{i_j}) = A_{i_1} \cdots A_{i_j}$, extended linearly over convergent series, with empty word mapped to I_n. Iterated integration gives $\Phi^A(X) = R_A(S(X))$.

For convergent tensor series F, G with $F_0 = G_0 = 1$, extend $\Delta_{r,n}$ by replacing the responses in (1) with $R_A(F), R_A(G)$. Write F_j, G_j for degree components. Declare standard word tensors orthonormal; their Hilbert–Schmidt (HS) norm is Euclidean norm of word coefficients. Define

$$\mathcal{B}_r(F, G) = \sum_{j \geq 1} r^j \, \|F_j - G_j\|_{\mathrm{HS}} \,. \tag{2}$$

We consider pairs for which this sum is finite. Degreewise Cauchy–Schwarz gives, for $|A| \leq r$,

$$\|R_A(F_j - G_j)\|_{\mathrm{op}} \leq r^j \, \|F_j - G_j\|_{\mathrm{HS}} \,, \qquad \Delta_{r,n}(F, G) \leq \mathcal{B}_r(F, G).$$

Thus $\mathcal{B}_r \leq \varepsilon$ certifies response distance at most ε. When $\mathcal{B}_r > \varepsilon$, this test leaves response accuracy undecided. We quantify how much this tensor measure can overstate response separation.

Comparison pairs. Degree-two approximations provide pairs with an explicit first discrepancy. Tensor multiplication concatenates words. For $U_0 = 0$, define $e^U_{\otimes} = \exp_{\otimes} U = \sum_{q \geq 0} U^{\otimes q}/q!$, with $U^{\otimes 0} = 1$; $\log_{\otimes}$ is its formal inverse on series with constant term one. Brackets are commutators: $[U, V] = U \otimes V - V \otimes U$ for tensors and $[A, B] = AB - BA$ for matrices.

For partition $\pi = (0 = t_0 < \cdots < t_m = 1)$, let $X_i = X|_{[t_{i-1}, t_i]}$ and $\ell_{i,j} = (\log_{\otimes} S(X_i))_j$. Store $\ell_i^{\leq 2} = \ell_{i,1} + \ell_{i,2}$: displacement and signed area in the plane. Exact signature and decoded series are

$$F = S(X), \qquad G_\pi = \prod_{i=1}^{m} \exp_{\otimes}(\ell_i^{\leq 2}).$$

Indices i, j denote interval and tensor degree; products follow time order. Local signatures use corresponding integration intervals.

Degree-two log-ODE decoding evaluates these retained terms in the system:

$$\Phi^A(X) = R_A(F), \qquad \widehat{\Phi}^A_\pi(X) = R_A(G_\pi) = \prod_i \exp(R_A(\ell_i^{\leq 2})).$$

One tuple A acts on every interval:

$$X \xrightarrow{\text{split } \pi} \underbrace{(\ell_i^{\leq 2})_{i=1}^m}_{\text{stored data}} \xrightarrow{\text{exponentiate and compose}} \widehat{\Phi}^A_\pi(X).$$

2 A planar distance-to-certificate ratio

First compare a path with its degree-two representation. Their response distance measures which discarded information bounded systems can detect. Scaling path amplitude isolates its first nonzero degree.

Fix $d = 2$ and partition π. Interval factors agree through degree two, so their first possible discrepancies add:

$$D_3 = (F - G_\pi)_3 = \sum_i \ell_{i,3}.$$

Define dilation $(\delta_s F)_j = s^j F_j$. Scaling X by $s > 0$ at coefficient budget r is equivalent to dilation by $t = sr$ at unit budget. Write

$$\Delta_n(t) = \Delta_{1,n}(\delta_t F, \delta_t G_\pi), \qquad \mathcal{B}(t) = \mathcal{B}_1(\delta_t F, \delta_t G_\pi).$$

Proposition 1. *If $D_3 \neq 0$, then*

$$\frac{\Delta_n(t)}{\mathcal{B}(t)} \longrightarrow \rho := \frac{2\sqrt{2}}{9} \quad (t \downarrow 0),$$

uniformly in response dimension $n \geq 2$.

HS norm counts word coefficients independently. Matrix products share one budget-constrained tuple, and cubic Lie terms become nested commutators. A driver rotation reduces any planar cubic Lie direction to a multiple of $[e_1, [e_1, e_2]]$. Its coefficients $1, -2, 1$ give HS norm $\sqrt{6}$. Böttcher–Wenzel [2] gives sharp response maximum $4/(3\sqrt{3})$, attained in dimension two. Their quotient is ρ, so leading certificate error exceeds response error by factor $1/\rho \approx 3.18$. Appendix A proves the gauge and tail bounds.

A finite-scale separation. For corner $0 \to e_1 \to e_1 + e_2$, store displacement and area in one interval. At unit budget,

$$\Phi^A(tX) = e^{tA_1} e^{tA_2}, \qquad \widehat{\Phi}^A_\pi(tX) = e^{t(A_1+A_2)+(t^2/2)[A_1,A_2]}.$$

At $t = 1/20$, rational enclosures prove, for every $n \geq 2$,

$$\Delta_n(t) < 1.934 \times 10^{-5} < 2 \times 10^{-5} < 3.608 \times 10^{-5} < \mathcal{B}(t). \tag{3}$$

These representations are within response distance 2×10^{-5}, while their tensor certificate exceeds that tolerance. Appendix A gives exact arithmetic.

A two-interval square loop can have $D_3 = 0$ and nonzero quartic error; partition refinement can make D_3 vanish. Mesh-uniform bounds and extension to rough inputs with roughness exponent $2 < p < 3$ remain open.

3 Area-chain distance separation at every amplitude

Our second comparison replaces an ordered product of area blocks by an exponential of their sum. Total area is preserved; order can affect higher-degree tensors and matrix responses. For $\ell \geq 3$, $d = \ell + 1$, set $U_i = [e_i, e_{i+1}]$, $1 \leq i \leq \ell$, and compare

$$F_\lambda = \prod_{i=1}^{\ell} e_\otimes^{\lambda U_i}, \qquad G_\lambda = e_\otimes^{\lambda \sum_i U_i}, \qquad \lambda > 0.$$

Here λ is area amplitude. Write $\Delta_n(\lambda) = \Delta_{1,n}(F_\lambda, G_\lambda)$ and $\mathcal{B}(\lambda) = \mathcal{B}_1(F_\lambda, G_\lambda)$.

Proposition 2. *For every $\ell \geq 3$, $n \geq 1$, and $\lambda > 0$,*

$$\Delta_n(\lambda) < \frac{\lambda^2}{\sqrt{2}} e^{\sqrt{2}\lambda} < \frac{e^{\sqrt{6}\lambda} - 1 - \sqrt{6}\lambda}{\sqrt{3}} \leq \mathcal{B}(\lambda). \tag{4}$$

Bounds are independent of chain length.

This separates the response and tensor measures for the stated pair at every amplitude. If F_λ itself approximates a BV path, add that initial response error by triangle inequality. Appendix B proves the bounds. Compression is one consequence: under fixed generic depth-two layout, replacing ℓ records by one reduces storage by factor ℓ at tolerances inside the proved gap.

Prior work and scope. The (ε, r)-representation paper supplies response criterion and identifies first discarded local log-signature error after (2.10) [1]. Proofs use Böttcher–Wenzel [2] and a two-factor commutator error bound [3]. Independent proof review and novelty confirmation remain pending.

A Planar gauge, tails and finite example

Let $Z = [e_1, e_2]$. Every planar cubic Lie element is $D = [w, Z]$, $w \in \mathbb{R}^2$, with $\|D\|_{\mathrm{HS}} = \sqrt{6}\,\|w\|_2$. Proper driver rotations preserve tuple ball and fix Z, reducing to $D = \|w\|_2\,[e_1, Z]$. Böttcher–Wenzel applied twice gives

$$\|[A_1, [A_1, A_2]]\|_{\mathrm{op}} \le 2\,\|A_1\|_F^2\,\|A_2\|_F \le \frac{4}{3\sqrt{3}} \quad (|A| \le 1).$$

Equality holds at

$$A_1 = \frac{1}{\sqrt{3}}\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \qquad A_2 = \frac{1}{\sqrt{3}}\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}.$$

Embedding in dimension $n \ge 2$ therefore proves $\sup_{|A|\le 1} \|R_A(D)\|_{\mathrm{op}} = \rho\,\|D\|_{\mathrm{HS}}$. One commutator step also gives $\sup_{|A|\le 1} \|R_A(\alpha Z)\|_{\mathrm{op}} \le |\alpha|/\sqrt{2} = \|\alpha Z\|_{\mathrm{HS}}/2$, $\alpha \in \mathbb{R}$.

Full-series tails. Put $L = \|X\|_{1\text{-var}}$, $h = \|D_3\|_{\mathrm{HS}}$, $a = \sum_i \|\ell_{i,1}\|_2$, and $b = \sum_i \|\ell_{i,2}\|_{\mathrm{HS}}$. Define

$$T_4(a, b; t) = e^{at+bt^2} - 1 - at - (a^2/2 + b)t^2 - (a^3/6 + ab)t^3,$$

$Q_R(t) = T_4(L, 0; t) + T_4(a, b/2; t)$, and $Q_H(t) = T_4(L, 0; t) + T_4(a, b; t)$. Then

$$|\Delta_n(t) - \rho h t^3| \le Q_R(t), \qquad ht^3 \le \mathcal{B}(t) \le ht^3 + Q_H(t). \tag{5}$$

Indeed, $\|S(X)_j\|_{\mathrm{HS}} \le L^j/j!$ gives majorant e^{Lt}. Tensor products give decoder majorant e^{at+bt^2}; area response bound improves this to $e^{at+(b/2)t^2}$. Removing degrees zero through three gives Q_R, Q_H. Cubic discrepancy is $\sum_i \ell_{i,3}$, since multiplication by nonconstant factors raises degree. Cubic gauge and triangle inequality give response bound in (5); degree-three certificate term gives its lower bound. Hence

$$\left|\frac{\Delta_n(t)}{\mathcal{B}(t)} - \rho\right| \le \frac{Q_R(t) + \rho Q_H(t)}{ht^3} = O(t).$$

Tail constants are independent of n, proving Proposition 1. In particular, $Q_R(t) < (1-\rho)ht^3$ implies $\Delta_n(t) < \mathcal{B}(t)$.

Rational corner certificate. For the one-interval corner,

$$D_3 = ([e_1, Z] - [e_2, Z])/12, \quad L = 2, \quad h = \sqrt{3}/6, \quad a = \sqrt{2}, \quad b = \sqrt{2}/2.$$

For rational $x \ge 0$, integer $J \ge 0$, and $x < J + 2$, put $P_J(x) = \sum_{j=0}^{J} x^j/j!$. Successive omitted-term ratios give

$$P_J(x) \le e^x \le P_J(x) + \frac{x^{J+1}}{(J+1)!}\frac{1}{1 - x/(J+2)}.$$

Use $J = 24$ and rational bounds, verified by integer squares,

$$\sqrt{2} < 707107/500000, \quad \sqrt{3} > 34641/20000, \quad \sqrt{6} < 244949/100000.$$

At $t = 1/20$, apply these to $\rho h t^3 + Q_R(t)$, using $\rho h = \sqrt{6}/27$ and nonnegative coefficients of T_4. The lower bound ht^3 for $\mathcal{B}(t)$ then proves (3) with exact fractions.

B All-scale chain proof

For $|A| \le 1$, put $C_i = [A_i, A_{i+1}]$, $a_i = \|A_i\|_F$, and $c_\ell = \cos(\pi/(\ell+2)) < 1$. The path-graph largest eigenvalue gives $\sum_{i=1}^{\ell} a_i a_{i+1} \le c_\ell \sum_i a_i^2 \le c_\ell$. Böttcher–Wenzel gives

$$M := \sup_{|A|\le 1} \sum_i \|C_i\|_{\mathrm{op}} \le \sqrt{2} c_\ell, \qquad K := \sup_{|A|\le 1} \sum_{i<j} \|[C_i, C_j]\|_{\mathrm{op}} \le \sqrt{2} c_\ell^2.$$

For the second bound, use $\|[C_i, C_j]\|_{\mathrm{op}} \le 2\sqrt{2}(a_i a_{i+1})(a_j a_{j+1})$, then sum pairs. For real square matrices P, Q, two-factor commutator bound [3] is

$$\left\| e^{\lambda P} e^{\lambda Q} - e^{\lambda(P+Q)} \right\|_{\mathrm{op}} \le \frac{\lambda^2}{2} \|[P, Q]\|_{\mathrm{op}} \, e^{\lambda(\|P\|_{\mathrm{op}} + \|Q\|_{\mathrm{op}})}.$$

Differentiate $e^{(\lambda-s)(P+Q)} e^{sP} e^{sQ}$ and use $[e^{sP}, Q] = \int_0^s e^{(s-u)P} [P, Q] e^{uP} \, du$. Induction over factors gives

$$\Delta_n(\lambda) \le \frac{\lambda^2 K}{2} e^{\lambda M} < \frac{\lambda^2}{\sqrt{2}} e^{\sqrt{2}\lambda}.$$

Write $F_\lambda - G_\lambda = \sum_{m \ge 2} \lambda^m D_{2m}$ and $H_m = m! \, \|D_{2m}\|_{\mathrm{HS}}$; tensor degree is $2m$. Blocks U_i are orthogonal with norm $\sqrt{2}$. Among words of length m over U_1, U_2, U_3, exactly $\binom{m+2}{2}$ are nondecreasing. Other words are absent from ordered product, and have coefficient $1/m!$ in merged exponential. Orthogonality gives

$$H_m^2 \ge 2^m \left(3^m - \binom{m+2}{2} \right) \ge 2^m 3^{m-1}, \quad m \ge 2.$$

The last inequality holds at $m = 2$ and follows by induction from $\binom{m+3}{2} / \binom{m+2}{2} = (m+3)/(m+1) < 3$. Summing $\mathcal{B}(\lambda) = \sum_{m \ge 2} \lambda^m H_m / m!$ proves last bound in (4).

For middle inequality, ratio of degree-m scalar coefficients is $L_m/\sqrt{2}$, where

$$L_m = \frac{4 \, 3^{(m-1)/2}}{m(m-1)}, \qquad \frac{L_{m+1}}{L_m} = \sqrt{3} \, \frac{m-1}{m+1}.$$

This ratio is below one for $m \le 3$ and above one for $m \ge 4$. Thus $\inf_{m \ge 2} L_m = L_4 = \sqrt{3} > \sqrt{2}$, proving strict coefficient comparison at every degree and Proposition 2.

References

[1] Ferrucci, Emilio; Perrée, Oliver; Lyons, Terry. *Concise (ε, r)-representations of a path.* 2026. https://arxiv.org/abs/2607.26281.

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[3] Iserles, Arieh; Kropielnicka, Karolina. *An elementary approach to splittings of unbounded operators.* 2024. https://arxiv.org/abs/2401.06635.